

DS 1 Experiment

Code:

```
#include <stdio.h>

void accept(int m, int n, int a[m][n]);
void display(int m, int n, int a[m][n]);
void add(int m, int n, int a[m][n], int b[m][n], int c[m][n]);
void subtract(int m, int n, int a[m][n], int b[m][n], int c[m][n]);
void multiply(int m, int n, int p, int a[m][n], int b[n][p], int c[m][p]);
void transpose(int m, int n, int a[m][n], int t[n][m]);

int main()
{
    int m, n, choice;

    printf("Enter rows and columns of Matrix A: ");
    scanf("%d%d", &m, &n);

    int a[m][n];

    printf("Enter Matrix A:\n");
    accept(m, n, a);

    do
    {
        printf("\n1.Addition\n2.Subtraction\n3.Multiplication\n4.Transpose\n5.Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
            {
                int b[m][n], sum[m][n];
                printf("Enter Matrix B:\n");
                accept(m, n, b);
                add(m, n, a, b, sum);
                printf("Addition:\n");
                display(m, n, sum);
                break;
            }

            case 2:
```

```

{
    int b[m][n], sub[m][n];
    printf("Enter Matrix B:\n");
    accept(m, n, b);
    subtract(m, n, a, b, sub);
    printf("Subtraction:\n");
    display(m, n, sub);
    break;
}

case 3:
{
    int p;
    printf("Enter number of columns of Matrix B: ");
    scanf("%d", &p);

    int b[n][p], mul[m][p];

    printf("Enter Matrix B (%d x %d):\n", n, p);
    accept(n, p, b);

    multiply(m, n, p, a, b, mul);

    printf("Multiplication:\n");
    display(m, p, mul);
    break;
}

case 4:
{
    int t[n][m];
    transpose(m, n, a, t);
    printf("Transpose:\n");
    display(n, m, t);
    break;
}

case 5:
    printf("Exiting...\n");
    break;

default:
    printf("Invalid choice\n");
}

```

```

    } while(choice != 5);

    return 0;
}

void accept(int m, int n, int a[m][n])
{
    for(int i = 0; i < m; i++)
        for(int j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
}

void display(int m, int n, int a[m][n])
{
    for(int i = 0; i < m; i++)
    {
        for(int j = 0; j < n; j++)
            printf("%d ", a[i][j]);
        printf("\n");
    }
}

void add(int m, int n, int a[m][n], int b[m][n], int c[m][n])
{
    for(int i = 0; i < m; i++)
        for(int j = 0; j < n; j++)
            c[i][j] = a[i][j] + b[i][j];
}

void subtract(int m, int n, int a[m][n], int b[m][n], int c[m][n])
{
    for(int i = 0; i < m; i++)
        for(int j = 0; j < n; j++)
            c[i][j] = a[i][j] - b[i][j];
}

void multiply(int m, int n, int p, int a[m][n], int b[n][p], int c[m][p])
{
    for(int i = 0; i < m; i++)
    {
        for(int j = 0; j < p; j++)
        {
            c[i][j] = 0;

```

```

        for(int k = 0; k < n; k++)
            c[i][j] += a[i][k] * b[k][j];
    }
}

void transpose(int m, int n, int a[m][n], int t[n][m])
{
    for(int i = 0; i < m; i++)
        for(int j = 0; j < n; j++)
            t[j][i] = a[i][j];
}

```

Output:

Enter rows and columns of Matrix A: 2

2

Enter Matrix A:

1

2

3

4

1.Addition

2.Subtraction

3.Multiplication

4.Transpose

5.Exit

Enter choice: 1

Enter Matrix B:

1

2

3

4

Addition:

2 4

6 8

1.Addition

2.Subtraction

3.Multiplication

4.Transpose

5.Exit

Enter choice: 2

Enter Matrix B:

1

2

3

4

Subtraction:

0 0

0 0

1.Addition

2.Subtraction

3.Multiplication

4.Transpose

5.Exit

Enter choice: 3

Enter number of columns of Matrix B: 2

Enter Matrix B (2 x 2):

1

2

3

4

Multiplication:

7 10

15 22

1.Addition

2.Subtraction

3.Multiplication

4.Transpose

5.Exit

Enter choice: 4

Transpose:

1 3

2 4

1.Addition

2.Subtraction

3.Multiplication

4.Transpose

5.Exit

Enter choice: 5

Exiting...